

SOC Day 2

Threat Detection and Investigation

1. Threat Detection

Threat detection means finding suspicious activity inside systems before attackers damage the company.

SOC analysts continuously monitor: Logs Servers User accounts Websites Network traffic Applications

Example: If someone tries hundreds of passwords on a server, SOC analysts should detect it quickly.

Main goal: Detect attacks early and stop attackers before damage increases.

```
# Failed login investigation
grep 'Failed password' /var/log/auth.log

# Successful logins
grep 'Accepted password' /var/log/auth.log

# Login history
last

# Running processes
ps aux

# Open ports
netstat -tulnp

# Active connections
ss -ant
```

2. Authentication Logs

Authentication logs store login related activity. These logs contain: Failed passwords Successful logins

SSH access sudo activity New account creation Important Linux file: /var/log/auth.log Example log: May

12 10:21:33 kali sshd[2211]: Failed password for root from 192.168.1.10 From this log analysts

understand: When attack happened Which account was targeted Which IP address attacked

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3. Brute Force Attack

Brute force attacks happen when attackers try many passwords continuously. Indicators: Many failed passwords Repeated attempts from same IP Successful login after many failures SOC analysts investigate: Source IP Target account Attack timing Successful compromise

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4. Password Spray Attack

Password spray attacks use one common password against many accounts. Example password: Welcome123 Target accounts: admin student manager Attackers use this method to avoid account

lockout. SOC analysts detect this by checking failed logins across many accounts.

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5. Web Log Analysis

Web logs store requests sent to websites. Important logs: /var/log/apache2/access.log /var/log/apache2/error.log SOC analysts detect: SQL injection XSS attacks Admin panel scanning Directory traversal Suspicious downloads

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6. SQL Injection

SQL injection attacks try to manipulate databases. Example: `?id=1' OR '1'=1` Attackers try to bypass login systems or access database information. SOC analysts search web logs for suspicious database related patterns.

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7. Network Investigation

SOC analysts monitor network traffic carefully. They investigate: Source IP Destination IP Ports Protocols Traffic timing Suspicious indicators: Unknown connections Port scanning Large outbound traffic Repeated failed connections

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8. SIEM Platforms

SIEM means Security Information and Event Management. SIEM platforms collect logs from many systems into one dashboard. Popular SIEM tools: Splunk ELK Stack Wazuh QRadar Benefits: Centralized monitoring Faster investigations Real time alerts Dashboard reporting

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9. Threat Hunting

Threat hunting means manually searching for hidden attackers. Threat hunters investigate: Hidden accounts Malware Credential theft Suspicious commands Data theft Good threat hunters follow small clues carefully.

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10. Malware Investigation

SOC analysts investigate malware activity carefully. They check: Running processes Unknown files Network communication Persistence methods File hashes Indicators: High CPU usage Unexpected services Unknown processes

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11. Incident Response

Incident response means handling attacks properly. Steps: Receive alert Collect logs Investigate attacker activity Contain attack Remove malware Recover systems Write report

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12. Timeline Analysis

SOC analysts create timelines to understand attacks step by step. Timelines help identify: Initial access
Privilege escalation Persistence Data theft Logs from different systems are combined together.

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13. Important Analyst Skills

SOC analysts should improve: Linux skills Networking Log analysis Threat detection Investigation skills
Documentation Strong analysts always investigate carefully and verify evidence.

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14. Final Notes

SOC work requires patience, observation, and investigation mindset. Students should practice: Linux commands Log analysis Network investigation Threat detection Daily practice improves cybersecurity skills.

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Important Security Tools

Tool	Purpose
Splunk	Enterprise log monitoring
ELK Stack	Open source SIEM
Wazuh	Endpoint monitoring
Wireshark	Packet analysis
tcpdump	Network capture

Created by Suman Kumar

HackShield Club IIT Patna

SOC Day 2 Notes